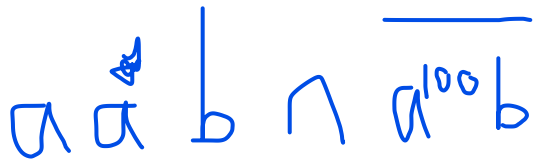
Theory of Computation

Exercise 4:

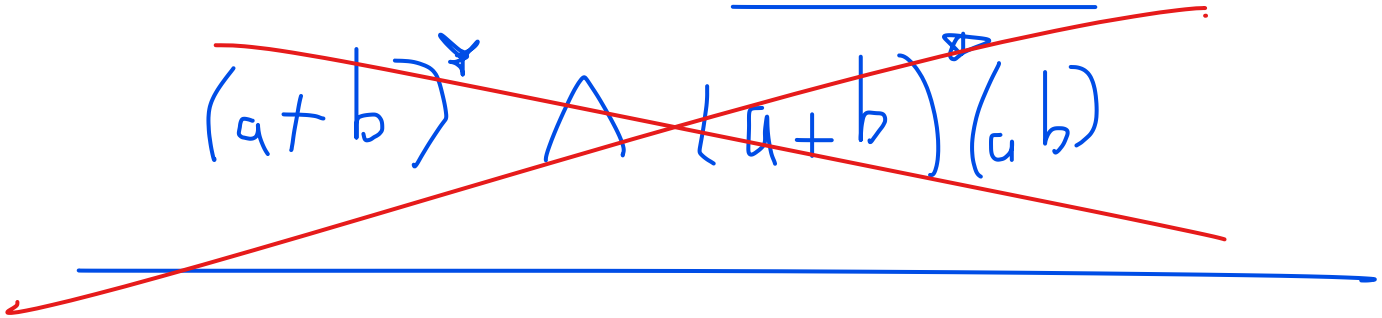
(Closure properties of Regular Language and Regular Expression)

1. Prove that the language $\{a^m b : m \geq 1 \; and \; m \neq 100\}$ is regular.



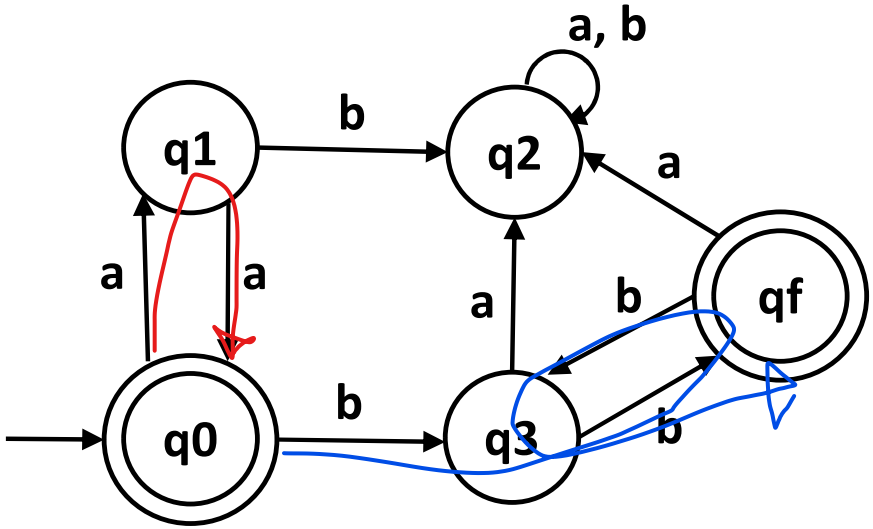
2. Find regular expression for the following language

$$L = \{ w \in \{a, b\}^* : w \text{ does not end with } ab \}$$



$(a+b)^*(aa+ba+bb)+\lambda+a+b$

*3. Find regular expression for the following DFA. (Homework 3)



(aa)* + (aa)*bb(bb)*

(a a)* + b b (bb)*